Input Token IDs

Token Embedding

× N

Adaptive Hybrid Block

same block structure, repeated N times

RMSNorm

normalized X (gate input)

Branch-specific projections

X_A X_S X_C

GQA Attention

Global information pathway

$A = GQA(X_A)$

Mamba Selective SSM

Efficient long-range pathway

$S = Mamba(X_S)$

Depthwise Conv1D

Local information pathway

$C = Conv1D(X_C)$

A S C

Adaptive 3-Way Fusion Gate

Linear → SiLU → Linear → Softmax → g_A g_S g_C

$g_A + g_S + g_C = 1$

$$H = g_A \cdot A + g_S \cdot S + g_C \cdot C$$

residual connection

H

$X' = X + H$

RMSNorm

Gated Feed-Forward Network

FFN

$Y = X' + FFN(RMSNorm(X'))$

Y → next Adaptive Hybrid Block

Final RMSNorm

LM Head

Next-Token Predictions

GQA · global pathway | Mamba SSM · long-range pathway | Conv1D · local pathway | Adaptive fusion